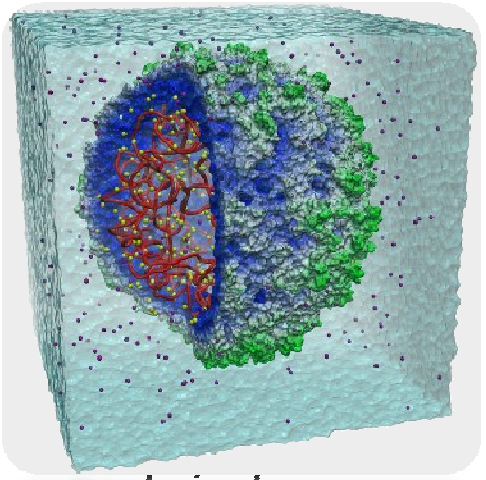
Concepção e Autoria Multimédia

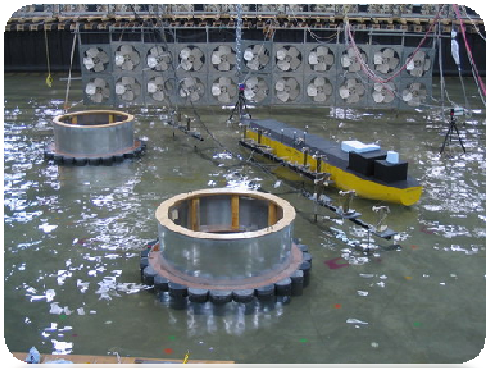
Carlos Vaz de Carvalho

Simulation and Modeling



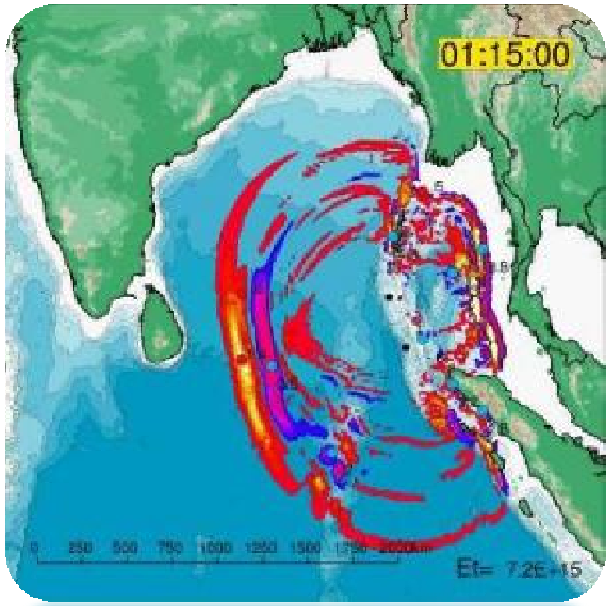
- Simulation is the imitation of some real **thing, state of affairs, or process**. The act of simulating something generally entails representing certain key **characteristics** or **behaviours** of a selected physical or abstract system.
- Simulation is used in many contexts, including the modeling of **natural systems** or **human systems** in order to gain insight into their functioning.
- Other contexts include simulation of technology for **performance optimization, safety engineering, testing, training** and **education**. Simulation can be used to show the eventual real effects of alternative conditions and courses of action.
- Key issues in simulation include acquisition of **valid source information** about the referent, selection of **key characteristics and behaviours**, the use of **simplifying approximations and assumptions** within the simulation, and **fidelity and validity of the simulation outcomes** - **modelling**

Simulation and Modelling



- **Physical simulation** refers to simulation in which physical objects are substitutes for the real thing. These physical objects are often chosen because they are smaller or cheaper than the actual object or system.
- **Interactive simulation** is a special kind of physical simulation, often referred to as a human in the loop simulation, in which physical simulations include human operators, such as in a flight simulator or a driving simulator.
- **Human in the loop simulations** can include a computer simulation as a so-called synthetic environment.

Computer Simulation



- **Computer simulation** has become a useful part of modeling many natural systems in **physics, chemistry** and **biology**, and human systems in **economics** and **social science** as well as in **engineering** to gain insight into the operation of those systems.
- A good example of the usefulness of using computers to simulate can be found in the field of **network traffic simulation**. In such simulations, the model behaviour will change each simulation according to the set of initial parameters assumed for the environment.
- Traditionally, the formal modeling of systems has been via a **mathematical model**, which attempts to find analytical solutions enabling the prediction of the behaviour of the system from a set of parameters and initial conditions.
- Computer simulation is often used as an adjunct to, or substitution for, modeling systems for which simple closed form analytic solutions are not possible. There are many different types of computer simulation, the common feature they all share is the attempt to generate a sample of representative scenarios for a model in which a complete enumeration of all possible states would be prohibitive or impossible.

Scientific Visualization

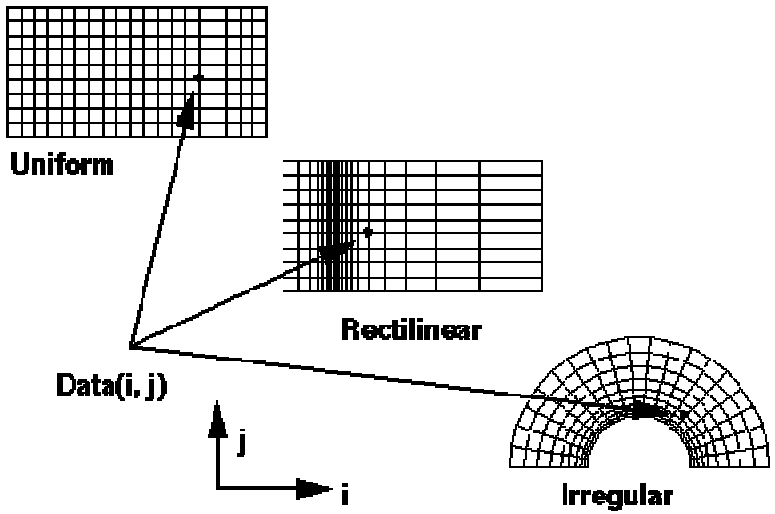
- Scientific visualization is the use of computer graphics to create **visual images which aid in understanding of complex**, often **massive** numerical representation of scientific concepts or results.
- Such numerical representations, or datasets, may be the **output of simulations** (e.g., fluid flow or molecular dynamics) or **empirical data** (e.g., recordings from geological, meteorological or astrophysical instruments).
- Scientific visualization is not an end in itself, but a component of many scientific tasks that typically combine interpretation and manipulation of scientific data and models. Scientists visualize data to look for patterns, features, relationships and anomalies--in other words, to aid understanding.

Scientific Visualization

- Daniel Thalmann (1990) presented scientific visualization as the new approach in the field of numerical simulation, which focuses on **basic geometric, animation and rendering**, as well as concrete applications in sciences and medicine.
- Underlying concepts presented are visualization, human perception, scientific methods, and the various aspects of data, such as acquisition, classification, storage and retrieval of data. The visualization techniques they have determined are **2-d, 3-d and multi-dimensional visualization techniques**, such as color transformations, glyphs for high dimensional data sets, visualization of gaseous and fluid information, volume rendering, isolines and isosurfaces, coloring, particle tracing, animation, techniques in virtual environments, and interactive steering.
- Information visualization is the study of the visual representation of large-scale collections of non-numerical information, such as files and lines of code in software systems library and bibliographic databases, networks of relations on the internet.
- Information visualization focused on the creation of approaches for conveying abstract information in intuitive ways. Visual representations and interaction techniques take advantage of the human eye's broad bandwidth pathway into the mind to allow users to see, explore, and understand large amounts of information at once.

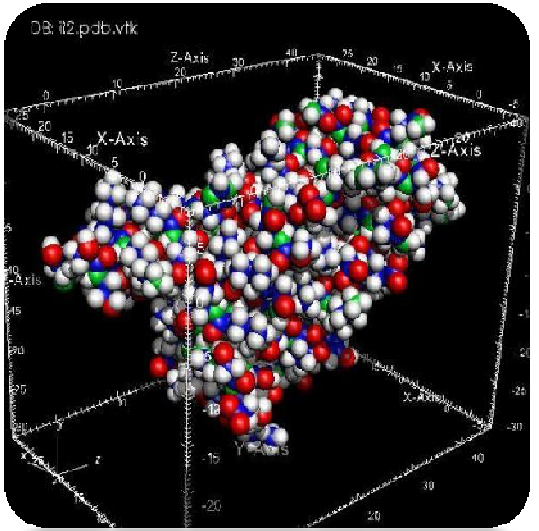
Scientific Visualization

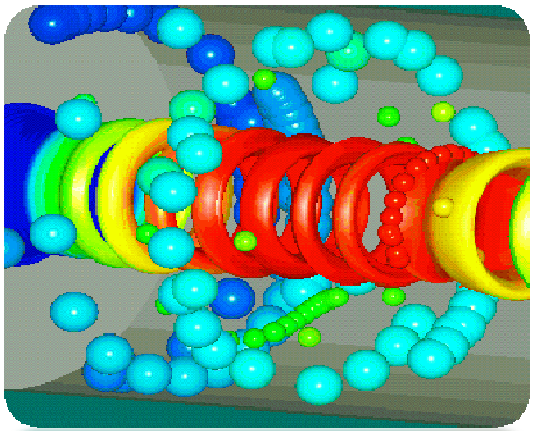
- Most data representing physical phenomena are sampled at discrete points on a grid or mesh.
- Data on grids that can be mapped into one or more rectangles/boxes is called structured data.
- Structured meshes can be uniform, rectilinear, or irregular.
- Multiple sub-grids can be combined to create multi-block grids.



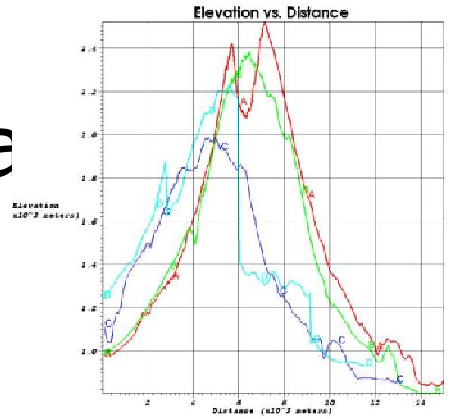
Scientific Visualization

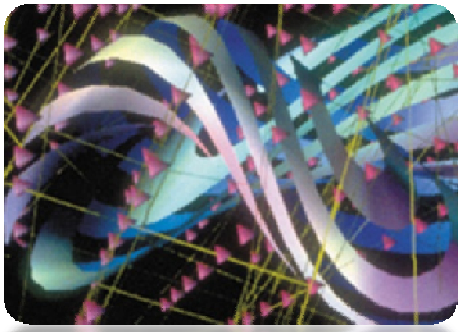
- Plots / Graphs
- Contour Lines/Isosurfaces
- Color Shading
- Glyphs
- Vector Arrows
- Particle Tracing
- Volume Visualization
- Animation

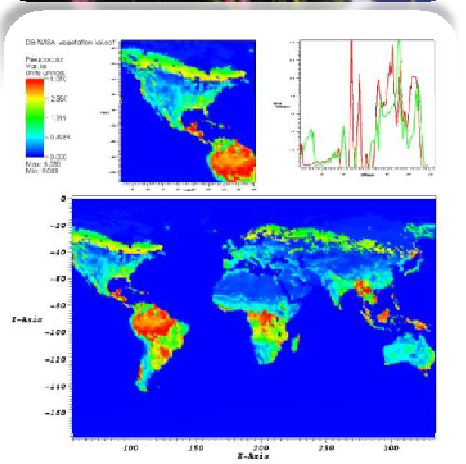


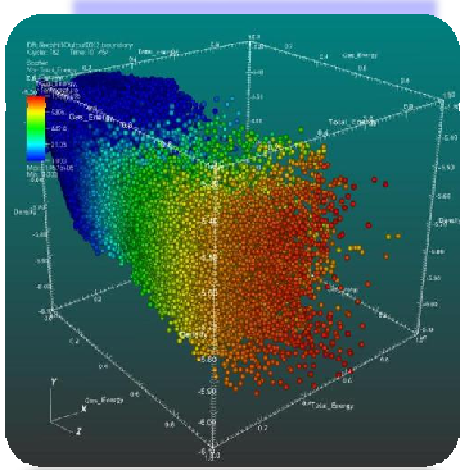


Scientific Visualiza Applications





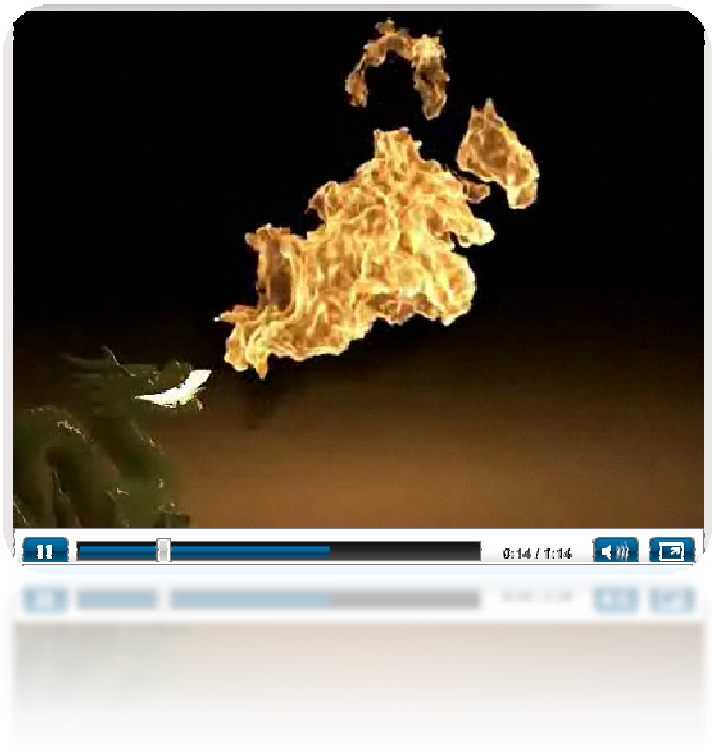




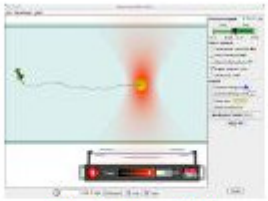
- **In the formal sciences**
- *Computer mapping of topographical surfaces*: Through computer mapping of topographical surfaces, mathematicians can test theories of how materials will change when stressed. The imaging is part of the work on the NSF-funded Electronic Visualization Laboratory at the University of Illinois at Chicago
- *Curve plots*: VisIt can plot curves from data read from files and it can be used to extract and plot curve data from higher dimensional datasets using lineout operators or queries. The curves in the featured image correspond to elevation data along lines drawn on DEM data and were created with the feature lineout capability.
- *Image annotations*: The featured plot shows Leaf Area Index (LAI), a measure of global vegetative matter, from a NetCDF dataset. The primary plot is the large plot at the bottom, which shows the LAI for the whole world.
- *Scatter plot*: VisIt's Scatter plot allows to visualize multivariate data of up to four dimensions. The Scatter plot takes multiple scalar variables and uses them for different axes in phase space.

Simulation Physics

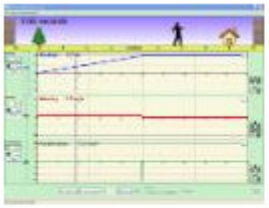
- Simple (educational) and complex physical simulations

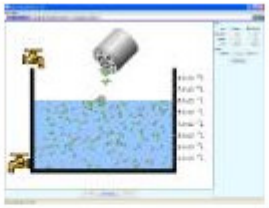


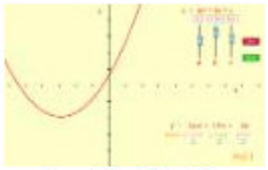

Circuit Construction Kit (DC Only)

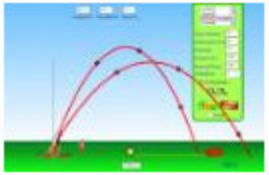

Stretching DNA


Glaciers

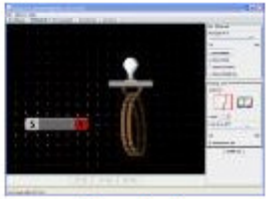

The Moving Man

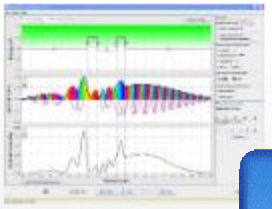

Salts & Solubility

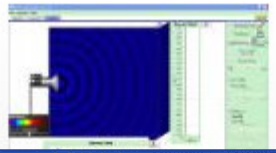

Equation Grapher


Projectile Motion


Lasers


Faraday's Electromagnetic Lab


Quantum Tunneling and Wave Packets

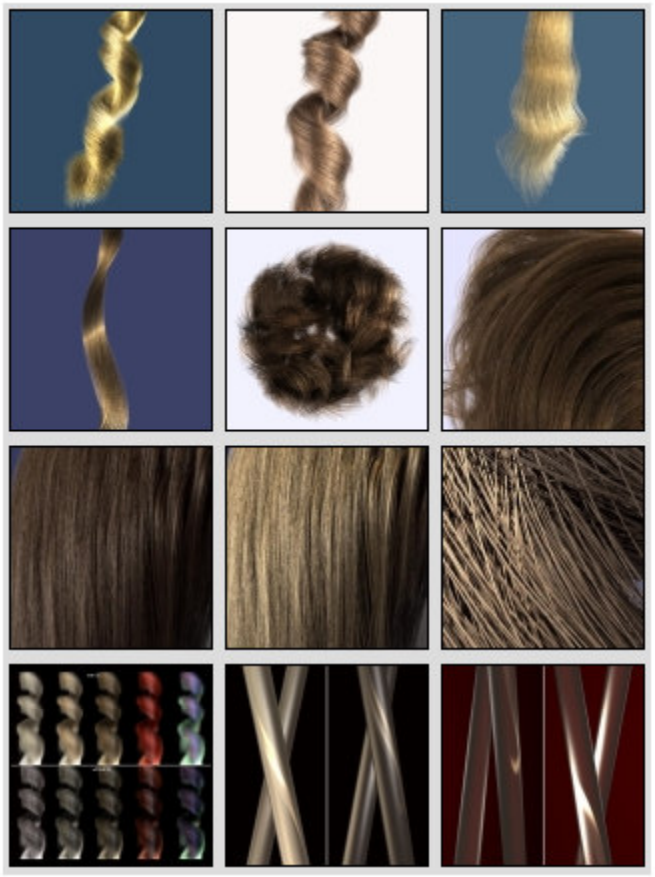



Radio Waves & Electromagnetic Fields



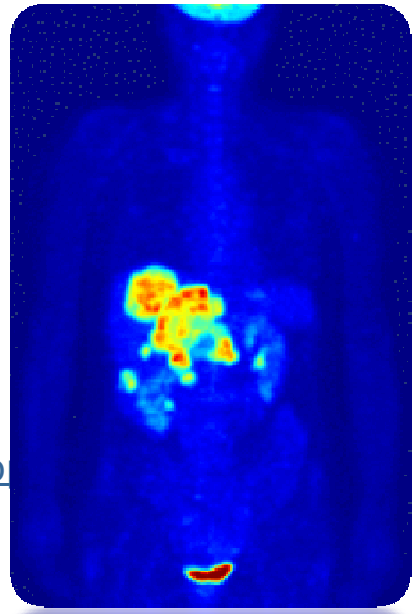
Hair Modelling

- Rendering hair implies that thousands or even millions of very thin strands have to be simulated. Surprisingly the complexity also depends on the hair color. For light colored hair multiple fiber scattering (global illumination) plays an important role for the right perception of the hair color and must not be neglected. "Based on a general and physically accurate model for light scattering from filaments (**B**idirectional **F**iber **S**cattering **D**istribution **F**unction - **BFSDF**) we developed techniques for photo-realistic rendering of hair and other fiber based geometry."
- University of Bonn (http://cg.cs.uni-bonn.de/project-pages/hairmodeling/)

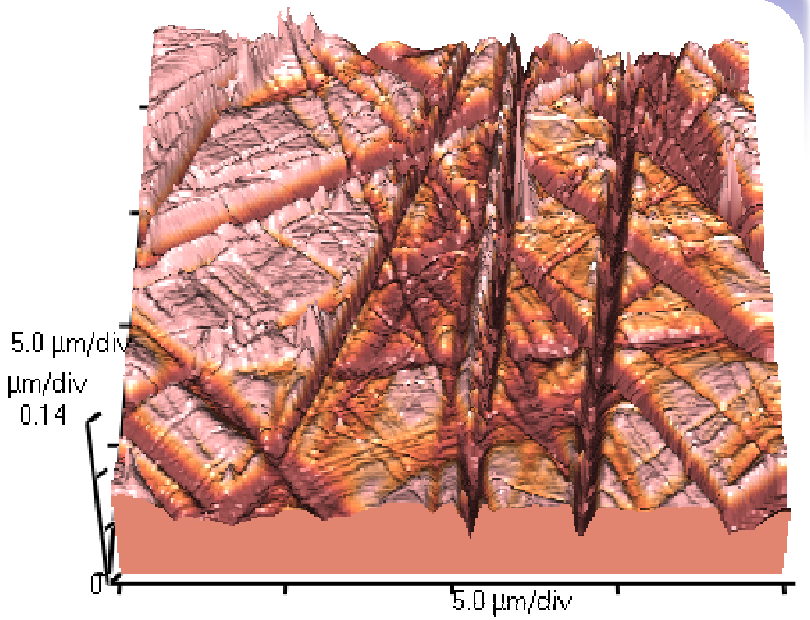


Scientific Visualization Applications

- Maximum intensity projection (MIP) of a whole body PET scan.
- Scientific visualization of Fluid Flow: Surface waves in water
- Topographic scan of a glass surface by an Atomic force microsco







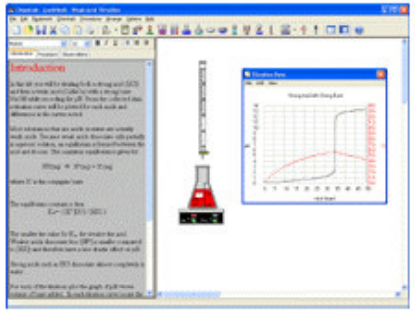
Simulation Chemistry



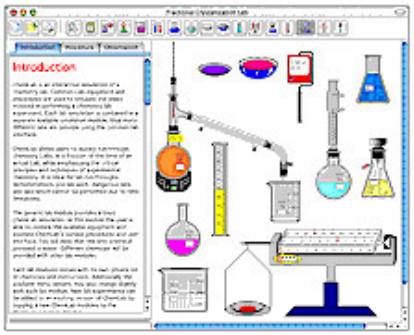
ChemLab Professional (LabWizard)



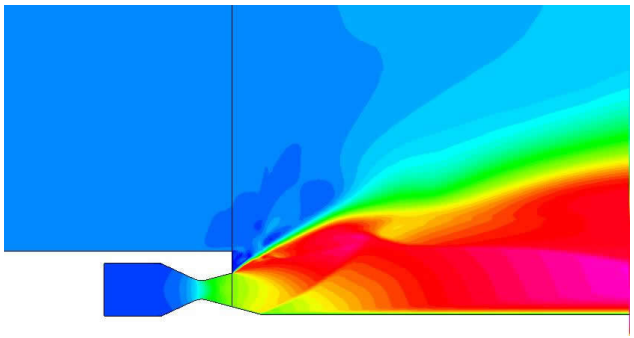
ChemLab Standard



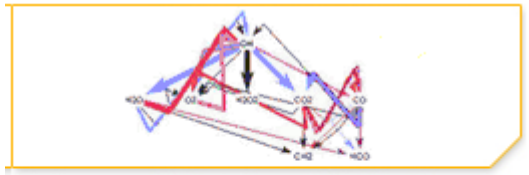
Titration Curve

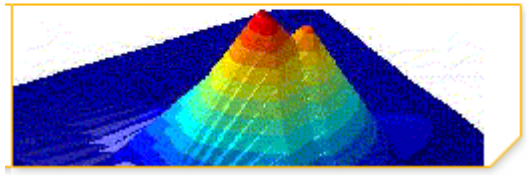


ChemLab - Mac OSX



NASA Glenn Research Center

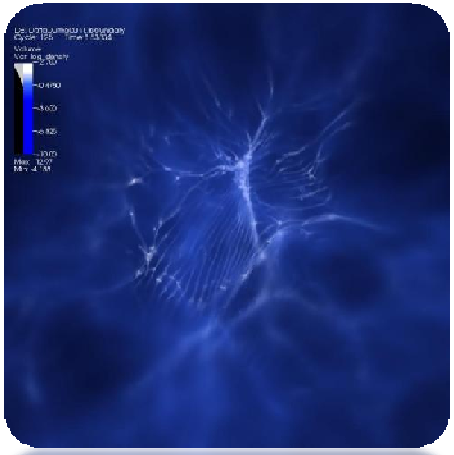


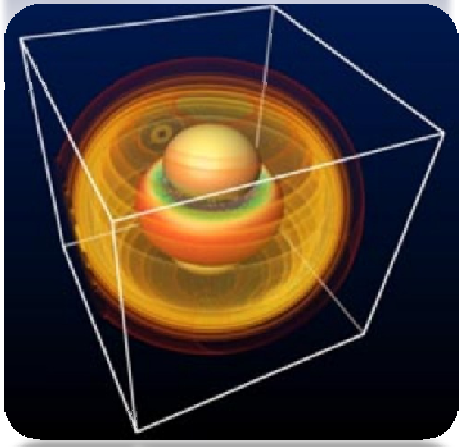


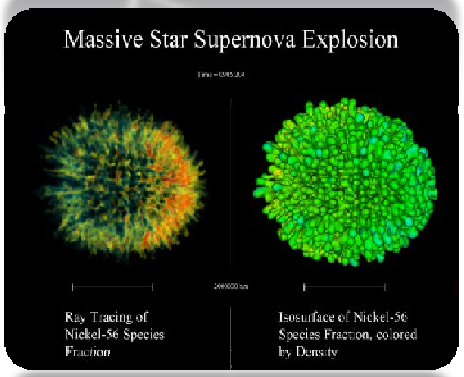
CHEMKIN-PRO

Scientific Visualization Applications

- *Star formation*: The featured plot is a Volume plot of the logarithm of gas/dust density in an Enzo star and galaxy simulation. Regions of high density are white while less dense regions are more blue and also more transparent.
- *Massive Star Supernovae Explosions*: In the image three Dimensional Radiation Hydrodynamics Calculations of Massive Star Supernovae Explosions The DJEHUTY stellar evolution code was used to calculate the explosion of SN 1987A model in three dimensions.
- *Molecular rendering*: VisIt's general plotting capabilities were used to create the molecular rendering shown in the featured visualization. The original data was taken from the Protein Data Bank and turned into a VTK file before rendering.







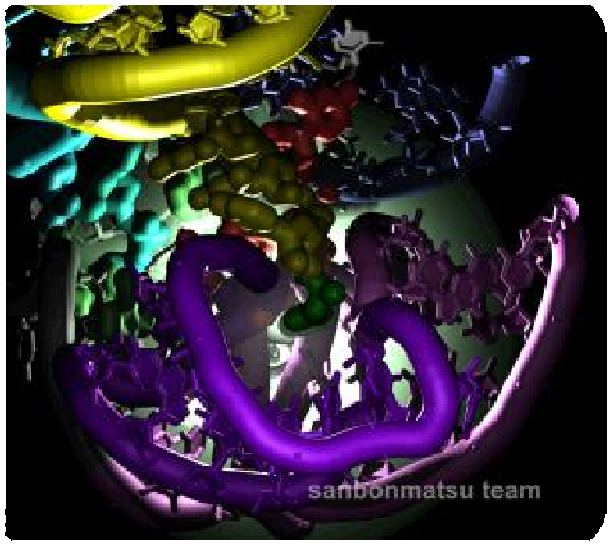
Scientific Visualization Applications

- **In the applied sciences**
- A smart gel is a material that gels in response to a specific physical property. For example, it may gel at a specific temperature or pressure.
- The mechanisms that create a gel in response to a given environment are not well understood. So developing this understanding is key to being able to create materials that gel at designated points.
- The potential for applications of smart gels is enormous. Since smart gels expand or contract in response to external stimuli, they could be useful in applications such as an artificial pancreas that releases insulin inside the body in response to high sugar level. Smart gels might someday be used to make exotic foods, cosmetics, medicines, and sensors.



Biology

- Researchers at Los Alamos National Laboratory have set a new world's record by performing the first million-atom computer simulation in biology. Using the "Q Machine" supercomputer, Los Alamos computer scientists have created a molecular simulation of the cell's protein-making structure, the ribosome. The project, simulating 2.64 million atoms in motion, is more than six times larger than any biological simulations performed to date….





Scientific Visualization Applications

- Bio Unbound: Stuff of Life is a collaborative project involving ACCAD, the Department of Design, the Department of Dance, and theDepartment of Entomology from OSU. The project is hosted by the Columbus Center Of Science and Industry (COSI) and represents an exhibit prototype for learning about DNA.
- The DNA multi-user touch screen workbench allows users to learn the concept of Complimentary Base Pairing by manipulating and connecting nucleotides by hand. Interacting with the wall projection next to the workbench, users engage their entire body in exploring the concept of super coiling.
- Accad – Advanced Computing Center for the Arts and Design (http://accad.osu.edu/research/scientific_visualization_htmls/biounbound.htm)



Social Science

- **Social simulation** is a research field that applies computational methods to study issues in the social sciences. The issues explored include problems in sociology, political science, economics, anthropology, geography, archaeology and linguistics





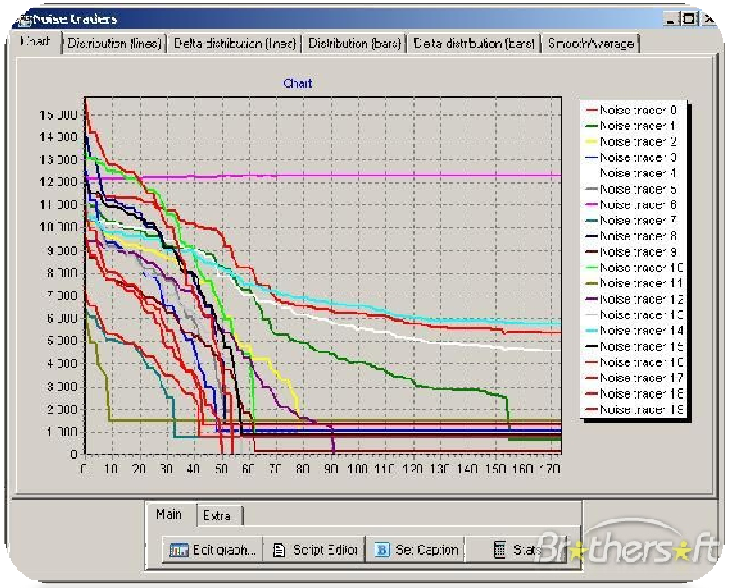
Education



- The "classroom of the future" will probably contain several kinds of simulators, in addition to textual and visual learning tools. The advanced student or postgraduate will have a more concise and comprehensive method of retraining.
- The classroom of the future will also form the basis of a clinical skills unit for continuing education of medical personnel; and in the same way that the use of periodic flight training assists airline pilots, this technology will assist practitioners throughout their career.
- The simulator will be more than a "living" textbook, it will become an integral a part of the practice of medicine. The simulator environment will also provide a standard platform for curriculum development in institutions of medical education.

Financial Simulation

- In finance, computer simulations are often used for scenario planning. Risk-adjusted net present value, for example, is computed from well-defined but not always known (or fixed) inputs. By imitating the performance of the project under evaluation, simulation can provide a distribution over a range of discount rates and other variables.





Sales process simulators

- Simulations are useful in modeling the flow of transactions through business processes, such as in the field of sales process engineering, to study and improve the flow of customer orders through various stages of completion (say, from an initial proposal for providing goods/services through order acceptance and installation).
- Such simulations can help predict the impact of how improvements in methods might impact variability, cost, labor time, and the quantity of transactions at various stages in the process.

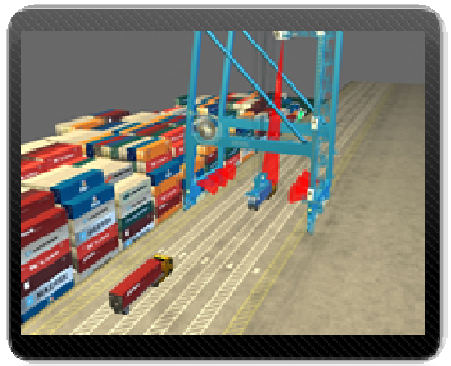
Employment simulation

- The student is 'employed' by Acumen Research to undertake a research project for a client, a large bank. In their office at Acumen (Figure 1) are various resources, including information on office procedures (principally on navigating in the simulation), books on research and statistics (electronic version of textbooks or guides), and a folder containing information on the project. Two general resources are available throughout the simulation, a phone which provides context sensitive help (e.g., hints on what to do next), and a notepad primarily for taking notes or storing information.
- University of Technology of Western Australia, http://www.ascilite.org.au/conferences/perth97/papers/Standen/Standen.html



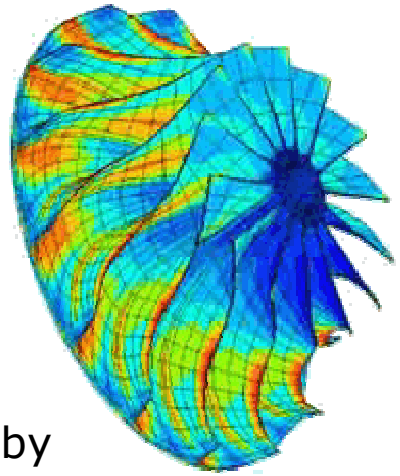


Simulation in Engineering

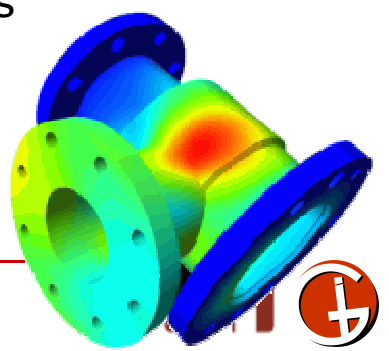


- *Simulation-Based Engineering Science (SBES)* is defined as the discipline that provides the **scientific and mathematical basis for the simulation of engineered systems**.
- Such systems range from microelectronic devices to automobiles, aircraft, and even the infrastructures of oilfields and cities.
- SBES fuses the knowledge and techniques of the traditional engineering fields—electrical, mechanical, civil, chemical, aerospace, nuclear, biomedical, and materials science—with the knowledge and techniques of fields like computer science, mathematics, and the physical and social sciences.

Simulation in Engineering

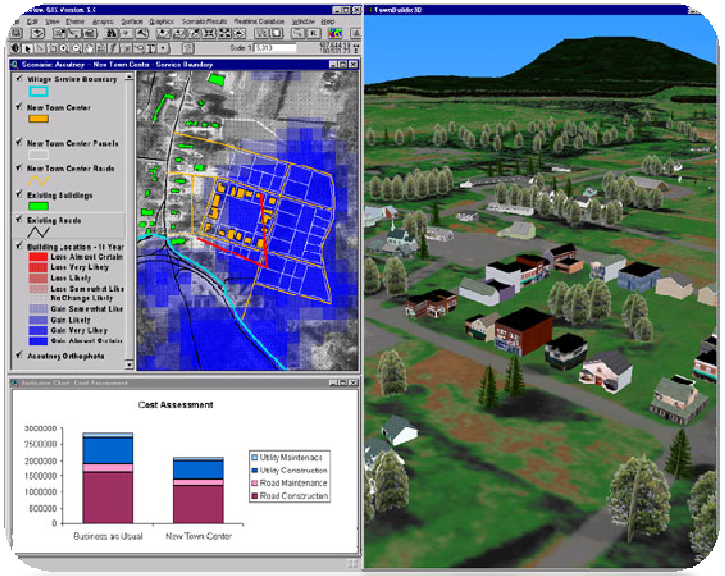


- For example in electrical engineering, delay lines may be used to simulate propagation delay and phase shift caused by an actual transmission line. Similarly, dummy loads may be used to simulate impedance without simulating propagation, and is used in situations where propagation is unwanted. A simulator may imitate only a few of the operations and functions of the unit it simulates.
- Most engineering simulations entail mathematical modeling and computer assisted investigation. There are many cases, however, where mathematical modeling is not reliable. Simulation of fluid dynamics problems often require both mathematical and physical simulations. In these cases the physical models require dynamic similitude. Physical and chemical simulations have also direct realistic uses, rather than research uses; in chemical engineering, for example, process simulations are used to give the process parameters immediately used for operating chemical plants, such as oil refineries.



Urban Simulation

- A city simulator can be a game but can also be a tool used by urban planners to understand how cities are likely to evolve in response to various policy decisions. UrbanSim (developed at the University of Washington), AnyLogic, ILUTE (developed at the University of Toronto) and Distrimobs (developed at the University of Bologna) are examples of modern, large-scale urban simulators designed for use by urban planners.

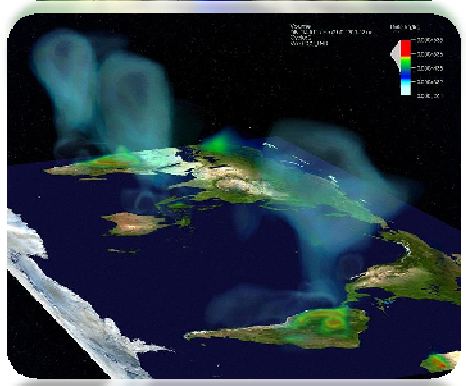


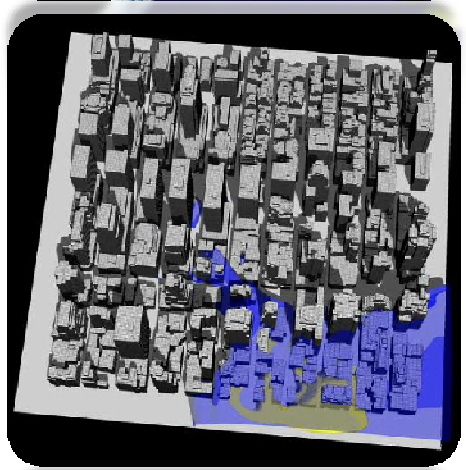


Scientific Visualization Applications

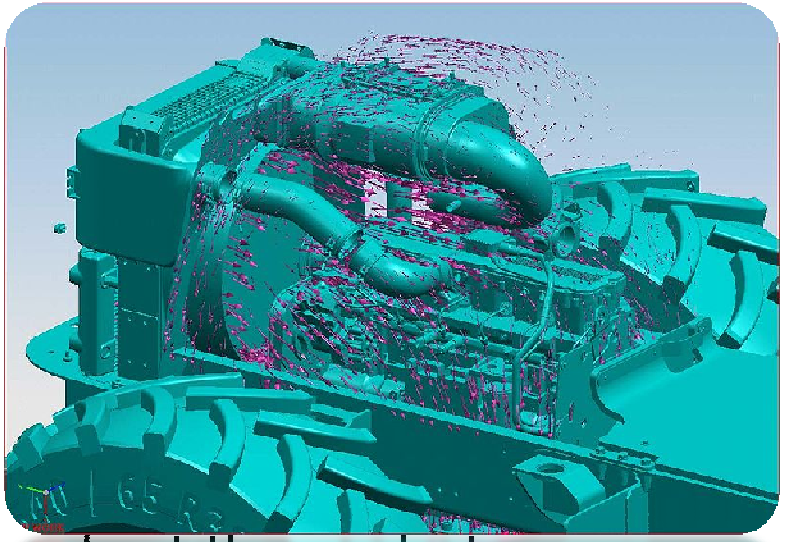
- *Terrain rendering*: VisIt can read several file formats common in the field of Geographic Information Systems (GIS), allowing one to plot raster data such as terrain data in visualizations. The featured image shows a plot of a DEM dataset containing mountainous areas near Dunsmuir, CA. Elevation lines are added to the plot to help delineate changes in elevation.
- *Tornado Simulation*: This image was created from data generated by a tornado simulation calculated on NCSA's IBM p690 computing cluster. High-definition television animations of the storm produced at NCSA were included in an episode of the PBS television series NOVA called "Hunt for the Supertwister." The tornado is shown by spheres that are colored according to pressure; orange and blue tubes represent the rising and falling airflow around the tornado.
- *Atmospheric Anomaly in Times Square* In the image the results from the SAMRAI simulation framework of an atmospheric anomaly in and around Times Square are visualized.







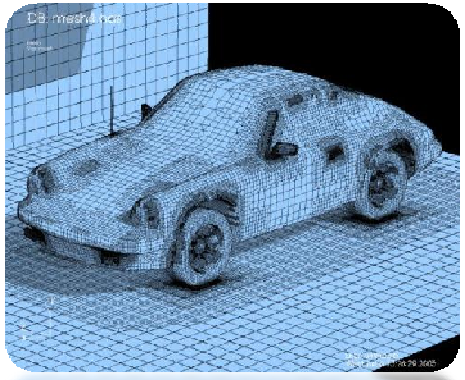
Product Lifecycle

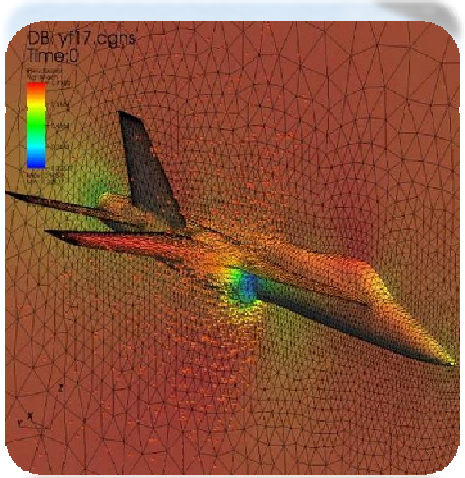


- Simulation solutions are being increasingly integrated with CAx (CAD, CAM, CAE....) solutions and processes. The use of simulation throughout the product lifecycle, especially at the earlier concept and design stages, has the potential of providing substantial benefits. These benefits range from direct cost issues such as reduced prototyping and shorter time-to-market, to better performing products and higher margins.
- The successful use of Simulation, early in the lifecycle, has been largely driven by increased integration of simulation tools with the entire CAD, CAM and PLM solution-set. Simulation solutions can now function across the extended enterprise in a multi-CAD environment, and include solutions for managing simulation data and processes and ensuring that simulation results are made part of the product lifecycle history.

Scientific Visualization Applications

- **In the applied sciences**
- *Porsche 911 model* (NASTRAN model): The featured plot contains a Mesh plot of a Porsche 911 model imported from a NASTRAN bulk data file. VisIt can read a limited subset of NASTRAN bulk data files, generally enough to import model geometry for visualization.
- *YF-17 aircraft Plot*: The featured image displays plots of a CGNS dataset representing a YF-17 jet aircraft. The dataset consists of an unstructured grid with solution. The image was created by using a pseudocolor plot of the dataset's Mach variable, a Mesh plot of the grid, and Vector plot of a slice through the Velocity field.
- *City rendering*: An ESRI shapefile containing a polygonal description of the building footprints was read in and then the polygons were resampled onto a rectilinear grid, which was extruded into the featured cityscape.







Flight simulators

- A flight simulator is used to train pilots on the ground. It permits a pilot to crash his simulated "aircraft" without being hurt. Flight simulators are often used to train pilots to operate aircraft in extremely hazardous situations, such as landings with no engines, or complete electrical or hydraulic failures. The most advanced simulators have high-fidelity visual systems and hydraulic motion systems. The simulator is normally cheaper to operate than a real trainer aircraft.



Marine simulators



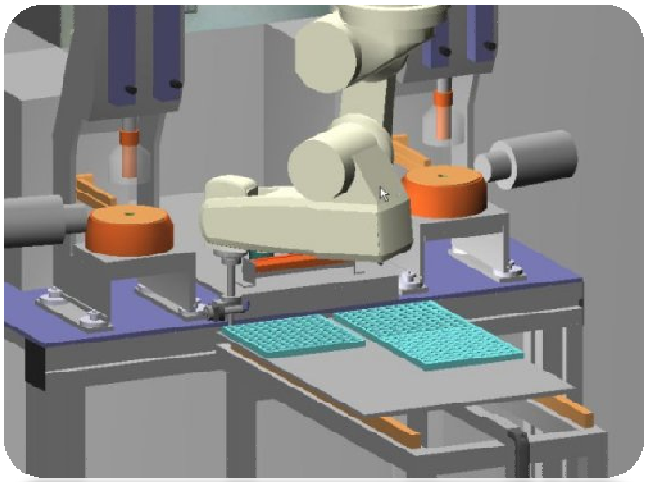
- Bearing resemblance to flight simulators, marine simulators train ships' personnel. The most common marine simulators include:
  - Ship's bridge simulators
  - Engine room simulators
  - Cargo handling simulators
  - Communication / GMDSS simulators
- Simulators like these are mostly used within maritime colleges, training institutions and navies. They often consist of a replication of a ships' bridge, with operating desk(s), and a number of screens on which the virtual surroundings are projected.

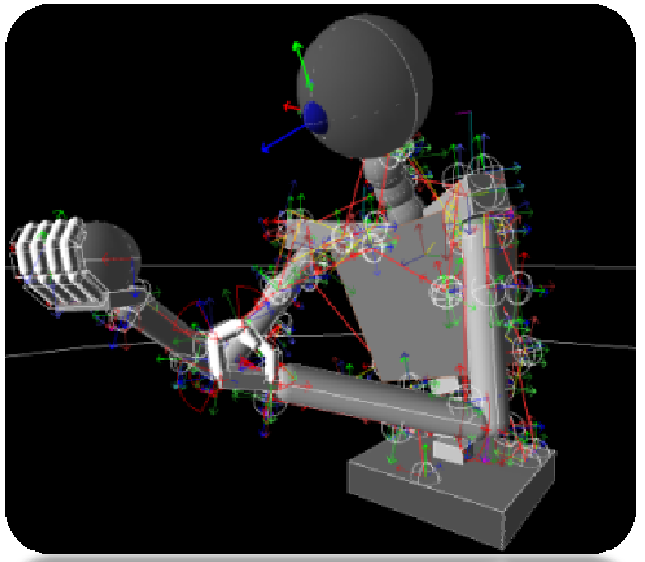
Military Simulations



- Military simulations, also known informally as war games, are models in which theories of warfare can be tested and refined without the need for actual hostilities. They exist in many different forms, with varying degrees of realism. In recent times, their scope has widened to include not only military but also political and social factors (for example, the NationLab series of strategic exercises in Latin America).

Robotics Simulators





- A robotics simulator is used to create embedded applications for a specific (or not) robot. In some cases, these applications can be transferred to the real robot (or rebuilt) without modifications.
- Robotics simulators allow reproducing situations that cannot be 'created' in the real world because of cost, time, or the 'uniqueness' of a resource. A simulator also allows fast robot prototyping. Many robot simulators feature physics engines to simulate a robot's dynamics.

Truck simulator

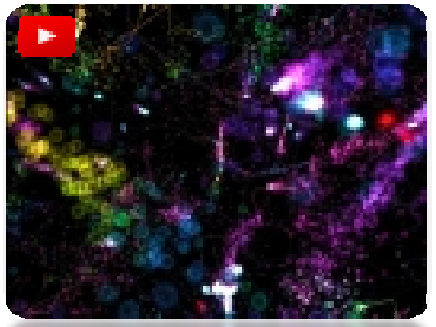


- A truck simulator provides an opportunity to reproduce the characteristics of real vehicles in a virtual environment. It replicates the external factors and conditions with which a vehicle interacts enabling a driver to feel as if they are sitting in the cab of their own vehicle. Scenarios and events are replicated with sufficient reality to ensure that drivers become fully immersed in the experience rather than simply viewing it as an educational programme.
- The simulator provides a constructive experience for the novice driver and enables more complex exercises to be undertaken by the more mature driver. For novice drivers, truck simulators provide an opportunity to begin their career by applying best practice. For mature drivers, simulation provides the ability to enhance good driving or to detect poor practice and to suggest the necessary steps for remedial action. For companies, it provides an opportunity to educate staff in the driving skills that achieve reduced maintenance costs, improved productivity and, most importantly, to ensure the safety of their actions in all possible situations.

Simulation and games

- Strategy games — both traditional and modern — may be viewed as simulations of abstracted decision-making for the purpose of training military and political leaders (see History of Go for an example of such a tradition, or Kriegsspiel for a more recent example).
- Many other video games are simulators of some kind. Such games can simulate various aspects of reality, from business, to government, to construction, to piloting vehicles (see above).

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